

[Courseware \(/courses/HumanitiesScience/StatLearning/Winter2014/courseware\)](/courses/HumanitiesScience/StatLearning/Winter2014/courseware)[Course Info \(/courses/HumanitiesScience/StatLearning/Winter2014/info\)](/courses/HumanitiesScience/StatLearning/Winter2014/info)[Discussion \(/courses/HumanitiesScience/StatLearning/Winter2014/discussion/forum\)](/courses/HumanitiesScience/StatLearning/Winter2014/discussion/forum)[Wiki \(/courses/HumanitiesScience/StatLearning/Winter2014/course_wiki\)](/courses/HumanitiesScience/StatLearning/Winter2014/course_wiki)[Progress \(/courses/HumanitiesScience/StatLearning/Winter2014/progress\)](/courses/HumanitiesScience/StatLearning/Winter2014/progress)[Lecture Slides \(pdf\) \(/courses/HumanitiesScience/StatLearning/Winter2014/ba1951b8f66c4cdca2fcaabdc91b792/\)](/courses/HumanitiesScience/StatLearning/Winter2014/ba1951b8f66c4cdca2fcaabdc91b792/)[R Sessions \(/courses/HumanitiesScience/StatLearning/Winter2014/0d68641c19484ef9aa6aeea03426dc68/\)](/courses/HumanitiesScience/StatLearning/Winter2014/0d68641c19484ef9aa6aeea03426dc68/)

3.3.R1 (1/1 point)

Suppose we are interested in learning about a relationship between X_1 and Y , which we would ideally like to interpret as causal.

True or False? The estimate $\hat{\beta}_1$ in a linear regression that controls for many variables (that is, a regression with many predictors in addition to X_1) is usually a more reliable measure of a causal relationship than $\hat{\beta}_1$ from a univariate regression on X_1 .

☐ True☒ False**EXPLANATION**

Adding lots of extra predictors to the model can just as easily muddy the interpretation of $\hat{\beta}_1$ as it can clarify it. One often reads in media reports of academic studies that "the investigators controlled for confounding variables," but be skeptical!

Causal inference is a difficult and slippery topic, which cannot be answered with observational data alone without additional assumptions.

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